

Ants on a Stick

Collision Invariance, Exit-Time Laws, and Probabilistic Structure

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August 2025

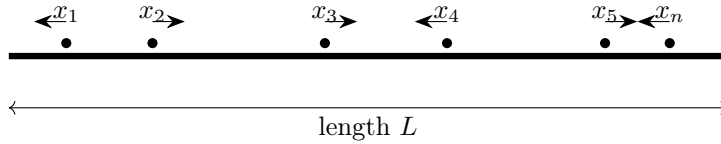


Figure 1: Ants modeled as identical point particles on an interval. Each moves at speed v in one of two directions, reverses direction at collisions, and exits at an endpoint.

Abstract

Identical equal-speed particles on a finite interval reverse velocities at collisions and leave when they reach an endpoint. The classical pass-through, or relabeling, representation replaces these interacting trajectories by noninteracting straight paths for every observable that ignores particle identities. Using this reduction, the finite ants-on-a-stick system is developed beyond the usual worst-case puzzle.

Completion time is an optimal $\Theta(n)$ statistic and the chronological exit sequence is comparison-optimal $\Theta(n \log n)$, even though the physical dynamics can contain $\Theta(n^2)$ collisions. With independent fair initial directions, the collision count is a binary-word inversion statistic with an exact q -binomial generating function, mean $n(n-1)/8$, and variance $n(n-1)(2n+5)/96$. For iid initial positions from a continuous common law, the directed-distance law is invariant under every joint direction law independent of the positions if and only if the common position law is symmetric about the midpoint. Uniform positions therefore identify the full exit process with uniform order statistics: normalized inter-exit spacings are Dirichlet, the number of particles remaining is a time-inhomogeneous pure-death process, and the final fixed-rank gaps have Gamma limits. For arbitrary continuous position laws, an exact completion-time distribution is obtained in terms of the initial direction count. The closed-form uniform model also yields an exact antithetic Monte Carlo benchmark, including correlation $-1/n$ and explicit variance ratios.

Keywords: one-dimensional hard-particle systems; order statistics; inversion statistics; extreme-value theory; pure-death processes; antithetic variates.

MSC 2020: 60E05 (primary); 60C05, 60G70, 60J27 (secondary).

1 Introduction and Model

The ants-on-a-stick problem is a familiar recreational model: identical ants move at common speed on a finite stick, reverse direction in pairwise collisions, and disappear at the endpoints. Its classical solution observes that a collision between two equal-speed ants is indistinguishable, at the level of unlabeled occupied positions, from allowing the trajectories to pass through and exchanging the particle labels. The puzzle and this reduction appear in recreational treatments such as Winkler [13] and the Harvey Mudd Math Fun Facts collection [12].

The same label-exchange principle is standard in one-dimensional equal-mass hard-particle systems. In the Jepsen gas, elastic collisions exchange velocities and thereby permit a mapping to free trajectories [9]. Related label-exchange and order-statistic representations are also used for single-file systems with

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absorbing boundaries [7, 10]. Collision statistics have been studied for more general freely expanding Jepsen gases with random velocities, where the collision count is tied to permutation inversions [11]. The two-velocity model considered here instead produces inversions in a binary word, connecting the collision count to standard random-word inversion enumerators [8].

The pass-through observation itself, the maximum of iid uniforms, standard order-statistic identities, and antithetic variates are established tools. This paper develops the finite-system structure that follows when those tools are applied consistently to the same model. The analysis isolates the exact midpoint-symmetry criterion for direction-law invariance, gives deterministic and comparison-model complexity bounds, records the binary collision-count law, identifies the complete uniform exit process through spacings and a remaining-particle Markov chain, derives fixed-rank terminal limits, extends completion times to arbitrary continuous position laws, and supplies exact Monte Carlo variance benchmarks.

Let the stick be the interval $[0, L]$, with $L > 0$. There are $n \geq 1$ point ants at distinct initial positions

$$0 \leq x_1, \dots, x_n \leq L,$$

and each has direction

$$d_i \in \{-1, +1\},$$

where -1 means left and $+1$ means right. Every ant moves at the same speed $v > 0$. Collisions are instantaneous and reverse both directions. An ant initialized at an endpoint and directed outward exits at time zero; one initialized at an endpoint and directed inward remains in the system and traverses the interval unless later relabeling changes which physical ant occupies that trajectory.

For the motivating two-foot instance,

$$L = 2 \text{ feet}, \quad v = 1 \text{ foot/minute},$$

so the natural time scale is $L/v = 2$ minutes. The analysis below is written for arbitrary L and v . Figure 1 summarizes the deterministic setup.

2 Collision Invariance and Exit Times

Lemma 2.1 (Collision invariance). *For two identical ants moving at equal speed on a line, collision-and-reversal and pass-through motion produce the same unordered set of positions after the meeting. They differ only in which label is attached to each outgoing trajectory.*

Proof. Immediately before a collision, one ant moves right and the other left. Reversing both velocities at the meeting point gives the same two outgoing geometric paths as allowing both ants to continue straight through the point. The paths are identical as an unlabeled set. $\square$

The local relabeling extends globally without a multiple-collision ambiguity.

Lemma 2.2 (No higher-order free intersection). *If the initial positions are distinct, no three free trajectories meet at one spacetime point at positive time.*

Proof. Before exit, every free trajectory has the form

$$y_i(t) = x_i + vd_it, \quad d_i \in \{-1, +1\}.$$

Among any three trajectories, two have the same slope. If those two met at the same time, equality of their positions would imply equality of their initial positions, contradicting distinctness. $\square$

Any simultaneous pairwise crossings therefore occur at different spatial locations and can be relabeled independently. This makes the pass-through representation exact for unlabeled occupied positions and exit times.

For each ant, define the directed distance to the endpoint it initially faces:

$$\delta_i = \begin{cases} x_i, & d_i = -1, \\ L - x_i, & d_i = +1. \end{cases} \quad (1)$$

Theorem 2.3 (Exit-time multiset). *Under collision-and-reversal dynamics, the unordered multiset of exit times is*

$$\left\{ \frac{\delta_1}{v}, \dots, \frac{\delta_n}{v} \right\}.$$

If $\delta_{(1)} \leq \dots \leq \delta_{(n)}$ are the ordered directed distances, then

$$T_{(k)} = \frac{\delta_{(k)}}{v},$$

and the system completion time is

$$T = \frac{1}{v} \max_{1 \leq i \leq n} \delta_i. \quad (2)$$

Proof. The pass-through representation has exactly the same unlabeled evolution as the physical system. Its i th free trajectory reaches the endpoint it faces after δ_i/v , so relabeling preserves the full multiset of exit times. $\square$

Remark 2.4 (Unlabeled versus labeled exits). The theorem determines the exit-time multiset, not the assignment of those times to the ants' initial identities. Physical ants never pass one another, so their surviving labels remain in their initial left-to-right order. After the free exit events are sorted, a left-end exit receives the leftmost surviving physical label and a right-end exit receives the rightmost surviving label; simultaneous exits at opposite endpoints remove both extremes. Thus identity-sensitive exit order is recovered by an induced relabeling permutation rather than by attaching the initial free-trajectory label to its exit time. The same distinction appears in Google Kick Start's 2022 problem "Ants on a Stick," which asks for the ants' original labels in exit order [5].

Sharp extremal bounds

Since $0 \leq \delta_i \leq L$,

$$T \leq \frac{L}{v}.$$

The bound is sharp. With endpoints allowed, an inward-moving ant at either endpoint has directed distance L , so

$$T_{\max} = \frac{L}{v}.$$

If all initial positions are required to lie in $(0, L)$, L/v remains the supremum but is not attained.

At the other endpoint, the infimum is zero. With distinct open-interval positions, no configuration attains zero, but configurations can approach it arbitrarily closely.

Corollary 2.5 (Lower extremum). *If all ants begin at distinct points in $(0, L)$, then the completion time has infimum 0, and the infimum is not attained.*

Proof. For every $\varepsilon > 0$, choose distinct $x_i \in (0, \min\{v\varepsilon, L\})$ and direct every ant left. Then $\delta_i = x_i < v\varepsilon$ for every i , so $T < \varepsilon$. Positive directed distances for interior starting points rule out $T = 0$. $\square$

For the two-foot, one-foot-per-minute version, the sharp upper bound is

$$\frac{L}{v} = 2 \text{ minutes} = 120 \text{ seconds}.$$

3 Algorithms and Collision Complexity

Equation (2) removes collision handling from the completion-time computation. A single scan computes the maximum directed distance, so completion time is available in $O(n)$ time and $O(1)$ auxiliary space.

Theorem 3.1 (Optimal completion-time complexity). *In the standard explicit-input model, exact completion time has worst-case complexity $\Theta(n)$.*

Proof. The one-pass maximum gives the upper bound. For the lower bound, suppose an algorithm leaves some ant's input unread. Holding every inspected input fixed, change that ant's directed distance from a small value to a value arbitrarily close to L . The completion time can change while the algorithm's observations do not. Hence every ant must be inspected in the worst case [1]. $\square$

The full ordered exit sequence requires more information. Computing all δ_i/v and comparison-sorting them takes $O(n \log n)$. This is optimal in the comparison model: with $v = 1$ and all ants directed left, arbitrary distinct keys $a_i \in (0, L)$ can be encoded as positions $x_i = a_i$, and the ordered exit times are exactly the sorted keys. Thus the ordered sequence has comparison complexity $\Theta(n \log n)$ [1].

How many collisions can occur?

Sort the initial positions so that $x_1 < \dots < x_n$. A physical collision occurs exactly when a right-moving free trajectory starting to the left crosses a left-moving free trajectory starting to the right. Therefore

$$C = \#\{(i, j) : i < j, d_i = +1, d_j = -1\}. \quad (3)$$

Thus C is the inversion count of the binary direction sequence.

If there are r right-moving and $n - r$ left-moving ants, then $C \leq r(n - r)$, with equality when all right-moving ants start to the left of all left-moving ants. Hence

$$C_{\max} = \left\lfloor \frac{n^2}{4} \right\rfloor. \quad (4)$$

The physical event sequence can therefore be quadratic even though its completion time is a one-pass statistic.

The binary-inversion representation gives the full collision-count distribution in a standard form. Let

$$R = \#\{i : d_i = +1\}$$

be the number of right-moving ants, and write $\begin{bmatrix} n \\ r \end{bmatrix}_q$ for the Gaussian binomial coefficient,

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \prod_{j=1}^r \frac{1 - q^{n-r+j}}{1 - q^j},$$

with the value at $q = 1$ understood by continuity.

Proposition 3.2 (Collision-count generating function). *Conditional on $R = r$ and a uniformly random direction word with r right-moving ants,*

$$\mathbb{E}[q^C \mid R = r] = \frac{\begin{bmatrix} n \\ r \end{bmatrix}_q}{\binom{n}{r}}. \quad (5)$$

If the directions are independent fair signs, then

$$\mathbb{E}[q^C] = 2^{-n} \sum_{r=0}^n \begin{bmatrix} n \\ r \end{bmatrix}_q. \quad (6)$$

Proof. For fixed r , equation (3) is the inversion number of a uniformly chosen binary word containing r symbols $+1$ and $n - r$ symbols -1 . The Gaussian binomial coefficient is the corresponding inversion enumerator [8]. Division by $\binom{n}{r}$ gives the conditional probability generating function. Under fair independent signs, $\mathbb{P}(R = r) = 2^{-n} \binom{n}{r}$, and averaging equation (5) over r yields equation (6). $\square$

The first two moments admit especially simple closed forms.

Theorem 3.3 (Collision moments under fair directions). *If $D_1, \dots, D_n$ are independent fair signs, then*

$$\mathbb{E}[C] = \frac{n(n-1)}{8}, \quad (7)$$

$$\text{Var}(C) = \frac{n(n-1)(2n+5)}{96}. \quad (8)$$

Consequently,

$$\frac{C}{n^2} \longrightarrow \frac{1}{8}$$

in L^2 and hence in probability.

Proof. Write

$$C = \sum_{i < j} I_{ij}, \quad I_{ij} = \mathbf{1}_{\{D_i=+1, D_j=-1\}}.$$

Each indicator has mean $1/4$ and variance $3/16$, giving equation (7). Indicators indexed by disjoint pairs are independent. For each triple $i < j < k$,

$$\text{Cov}(I_{ij}, I_{ik}) = \frac{1}{16}, \quad \text{Cov}(I_{ij}, I_{jk}) = -\frac{1}{16}, \quad \text{Cov}(I_{ik}, I_{jk}) = \frac{1}{16}.$$

Their sum is $1/16$. Therefore

$$\text{Var}(C) = \binom{n}{2} \frac{3}{16} + 2 \binom{n}{3} \frac{1}{16} = \frac{n(n-1)(2n+5)}{96}.$$

After division by n^4 , the variance tends to zero while $\mathbb{E}[C]/n^2 \rightarrow 1/8$, proving the final claim. $\square$

4 Random Initial Conditions and Reflection Symmetry

The deterministic reduction does not itself require random initial conditions. Randomness enters only through the initial configuration. The most useful structural question is therefore not which direction distribution is convenient, but when changing directions leaves the directed-distance law unchanged.

Let $X_1, \dots, X_n$ be iid from a continuous distribution on $[0, L]$, and let the direction vector $D = (D_1, \dots, D_n)$ have any probability law on $\{-1, +1\}^n$ that is independent of $X = (X_1, \dots, X_n)$. The directions may be biased and may be mutually dependent.

Theorem 4.1 (Exact reflection criterion). *The directed exit distances $\Delta_1, \dots, \Delta_n$ are iid with the same law as the initial positions for every joint direction law independent of the positions if and only if the common position law is symmetric about the midpoint:*

$$X_i \stackrel{d}{=} L - X_i.$$

Proof. For sufficiency, condition on any fixed direction vector $D = d$. Each coordinate Δ_i is then either X_i or $L - X_i$. Midpoint symmetry makes both choices equal in distribution, and deterministic coordinatewise reflections preserve independence because they act separately on the iid coordinates. Thus the conditional joint law of $(\Delta_1, \dots, \Delta_n)$ is the same iid product law for every d , and averaging over the arbitrary law of D leaves it unchanged.

For necessity, choose the deterministic all-left direction vector and the deterministic all-right direction vector. In the first case each directed distance has the law of X_i ; in the second it has the law of $L - X_i$. Invariance for every direction law therefore forces $X_i \stackrel{d}{=} L - X_i$. $\square$

Uniform positions are the canonical special case and give the strongest closed forms:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Uniform}(0, L) \implies \Delta_1, \dots, \Delta_n \stackrel{\text{iid}}{\sim} \text{Uniform}(0, L).$$

The fair-independent direction model is therefore sufficient but not necessary for any of the uniform exit-time results below.

Exit order, completion time, and covariance

Let $\Delta_{(1)} \leq \dots \leq \Delta_{(n)}$ be the order statistics. Standard uniform order-statistic theory gives [2]

$$\frac{vT_{(k)}}{L} = \frac{\Delta_{(k)}}{L} \sim \text{Beta}(k, n+1-k). \quad (9)$$

Hence

$$\mathbb{E}[T_{(k)}] = \frac{k}{n+1} \frac{L}{v}. \quad (10)$$

For $1 \leq i \leq j \leq n$, the joint covariance structure is

$$\text{Cov}(T_{(i)}, T_{(j)}) = \left(\frac{L}{v}\right)^2 \frac{i(n-j+1)}{(n+1)^2(n+2)}. \quad (11)$$

For $T = T_{(n)}$ and $0 \leq t \leq L/v$,

$$F_T(t) = \mathbb{P}(T \leq t) = \left(\frac{vt}{L}\right)^n, \quad (12)$$

with density

$$f_T(t) = n \left(\frac{v}{L}\right)^n t^{n-1}.$$

For every real $k > 0$,

$$\mathbb{E}[T^k] = \frac{n}{n+k} \left(\frac{L}{v}\right)^k. \quad (13)$$

Therefore

$$\mathbb{E}[T] = \frac{n}{n+1} \frac{L}{v}, \quad (14)$$

$$\text{Var}(T) = \frac{n}{(n+1)^2(n+2)} \left(\frac{L}{v}\right)^2, \quad (15)$$

$$Q_q(T) = \frac{L}{v} q^{1/n}, \quad 0 < q < 1. \quad (16)$$

For the two-foot classroom scale, $\mathbb{E}[T] = 2n/(n+1)$ minutes; this is $20/11 \approx 1.818$ minutes for $n = 10$ and $100/51 \approx 1.961$ minutes for $n = 50$.

The full spacing law

The individual Beta laws are marginals of a stronger joint result. Put

$$U_{(k)} = \frac{vT_{(k)}}{L}, \quad U_{(0)} = 0, \quad U_{(n+1)} = 1,$$

and define the $n+1$ spacings

$$S_k = U_{(k)} - U_{(k-1)}, \quad k = 1, \dots, n+1.$$

Theorem 4.2 (Uniform exit-time spacings). *The spacing vector satisfies*

$$(S_1, \dots, S_{n+1}) \sim \text{Dirichlet}(1, \dots, 1). \quad (17)$$

Consequently,

$$\mathbb{E}[S_k] = \frac{1}{n+1}, \quad \text{Var}(S_k) = \frac{n}{(n+1)^2(n+2)}, \quad (18)$$

and, for $i \neq j$,

$$\text{Cov}(S_i, S_j) = -\frac{1}{(n+1)^2(n+2)}. \quad (19)$$

In physical time, set $T_{(0)} = 0$ and $T_{(n+1)} = L/v$. Then every consecutive expected gap, including the initial and terminal endpoint gaps, is

$$\mathbb{E}[T_{(k)} - T_{(k-1)}] = \frac{L}{v(n+1)}, \quad k = 1, \dots, n+1. \quad (20)$$

Proof. The joint density of the ordered uniforms $(U_{(1)}, \dots, U_{(n)})$ is the constant $n!$ on the simplex

$$0 < u_1 < \dots < u_n < 1.$$

The change of variables from ordered locations to spacings has unit Jacobian and maps this simplex to

$$s_k \geq 0, \quad \sum_{k=1}^{n+1} s_k = 1.$$

The resulting constant density is exactly the $\text{Dirichlet}(1, \dots, 1)$ density. Its standard moments give equations (18)–(19), and multiplication by L/v gives equation (20). Equation (11) follows as well because each $U_{(k)}$ is a partial sum of the spacings [2]. $\square$

The mean completion-time shortfall from the sharp bound is therefore exactly one expected spacing:

$$\frac{L}{v} - \mathbb{E}[T] = \frac{L}{v(n+1)}. \quad (21)$$

For fixed n , repeated independent experiments recover the same expectation empirically. If $T_1, T_2, \dots$ are iid completion times from the uniform model, then boundedness and the strong law of large numbers give [6]

$$\frac{1}{m} \sum_{j=1}^m T_j \longrightarrow \frac{n}{n+1} \frac{L}{v} \quad \text{almost surely as } m \rightarrow \infty.$$

This repeated-trial limit keeps n fixed. It is distinct from the many-ant limit $\mathbb{E}[T] \rightarrow L/v$ as $n \rightarrow \infty$, whose exact gap is equation (21).

The ants-remaining process

At time t , define

$$R(t) = \#\{i : \Delta_i > vt\}.$$

For $0 \leq t \leq L/v$,

$$R(t) \sim \text{Binomial}\left(n, 1 - \frac{vt}{L}\right), \quad (22)$$

so

$$\mathbb{E}[R(t)] = n \left(1 - \frac{vt}{L}\right), \quad (23)$$

$$\text{Var}(R(t)) = n \left(1 - \frac{vt}{L}\right) \frac{vt}{L}. \quad (24)$$

The completion event is exactly $\{R(t) = 0\}$, which recovers equation (12).

The marginal binomial law is only part of the process description.

Proposition 4.3 (Transition law for ants remaining). *For $0 \leq s < t < L/v$,*

$$R(t) \mid R(s) = r \sim \text{Binomial}\left(r, \frac{L - vt}{L - vs}\right). \quad (25)$$

Thus $R(t)$ is a time-inhomogeneous pure-death Markov process. Its instantaneous transition rate from r to $r - 1$ is

$$q_{r,r-1}(t) = \frac{rv}{L - vt}. \quad (26)$$

Proof. Under the uniform model, the individual pass-through exit times are iid $\text{Uniform}(0, L/v)$. Conditional on an ant still being present at time s , its exit time is uniform on $(s, L/v)$. Therefore a surviving ant remains present at time t with probability

$$\frac{L/v - t}{L/v - s} = \frac{L - vt}{L - vs}.$$

Conditional on the r survivors at time s , these residual exit times remain independent, giving equation (25). For $h \downarrow 0$, the one-ant conditional exit probability over $(t, t + h]$ is $vh/(L - vt) + o(h)$; multiplying by r gives equation (26). $\square$

Terminal order statistics and extreme values

The usual completion-time limit is the first member of a fuller terminal hierarchy. For a fixed positive integer r , let $T_{(n-r+1)}$ be the r th-to-last exit time.

Theorem 4.4 (Gamma terminal-gap hierarchy). *For every fixed $r \geq 1$,*

$$n \left(1 - \frac{vT_{(n-r+1)}}{L}\right) \Rightarrow \text{Gamma}(r, 1), \quad (27)$$

where the Gamma distribution is parameterized by shape r and rate 1.

Proof. Fix $x \geq 0$, and let $N_n(x)$ count normalized exit times in $(1 - x/n, 1]$. Then

$$N_n(x) \sim \text{Binomial}\left(n, \frac{x}{n}\right) \Rightarrow \text{Poisson}(x).$$

The event

$$n(1 - U_{(n-r+1)}) > x$$

is equivalent to having at most $r - 1$ observations in $(1 - x/n, 1]$. Hence

$$\mathbb{P}(n(1 - U_{(n-r+1)}) > x) \longrightarrow e^{-x} \sum_{j=0}^{r-1} \frac{x^j}{j!},$$

which is the survival function of a $\text{Gamma}(r, 1)$ random variable. $\square$

For $r = 1$, equation (27) reduces to

$$n\left(1 - \frac{vT}{L}\right) \Rightarrow \text{Exponential}(1), \quad (28)$$

so the gap between completion and the upper endpoint is of order $1/n$. Equivalently, with $M_n = vT/L$ and $Z_n = n(M_n - 1) \leq 0$,

$$\mathbb{P}(Z_n \leq z) \longrightarrow H(z) = \begin{cases} e^z, & z \leq 0, \\ 1, & z > 0. \end{cases}$$

This is a Type III (reverse-Weibull) extreme-value law with generalized-extreme-value shape index $\xi = -1$; the uniform law lies in the Weibull maximum domain of attraction [3].

5 Beyond Uniform Positions

The reflection theorem isolates the special role of symmetric position laws. Without midpoint symmetry, the direction vector can change the completion-time distribution. The dependence can still be written exactly.

Suppose $X_1, \dots, X_n$ are iid with continuous CDF F on $[0, L]$, and let D have an arbitrary joint law independent of the positions. Define

$$K = \#\{i : D_i = -1\},$$

the number initially directed left, and for $0 \leq y \leq L$ write

$$a(y) = F(y), \quad b(y) = 1 - F(L - y).$$

Theorem 5.1 (General completion-time law). *For $0 \leq t \leq L/v$,*

$$\mathbb{P}(T \leq t) = \mathbb{E}[a(vt)^K b(vt)^{n-K}]. \quad (29)$$

Consequently,

$$\mathbb{E}[T] = \frac{1}{v} \int_0^L \{1 - \mathbb{E}[a(y)^K b(y)^{n-K}]\} dy. \quad (30)$$

Proof. Condition on the full direction vector. If $K = k$, then k directed distances have the form X_i and the remaining $n - k$ have the form $L - X_i$. Since the positions are independent,

$$\mathbb{P}(T \leq t \mid D) = a(vt)^K b(vt)^{n-K}.$$

Averaging over the direction law gives equation (29). The completion time is nonnegative and bounded by L/v , so applying the tail-integral identity and changing variables $y = vt$ gives equation (30). $\square$

Two useful special cases fall out immediately. If the directions are iid with

$$\mathbb{P}(D_i = -1) = p, \quad \mathbb{P}(D_i = +1) = 1 - p,$$

then $K \sim \text{Binomial}(n, p)$ and equation (29) becomes

$$\mathbb{P}(T \leq t) = G(vt)^n, \quad G(y) = pF(y) + (1 - p)[1 - F(L - y)]. \quad (31)$$

If instead F is midpoint-symmetric, then $a(y) = b(y)$ and equation (29) reduces to $F(vt)^n$ for *every* direction law independent of the positions, in agreement with the stronger joint reflection-invariance theorem. The uniform model is the case $F(y) = y/L$.

6 Monte Carlo and Antithetic Variance Reduction

The closed-form uniform model is useful as a simulation benchmark precisely because the target is known. Normalize

$$U_i = \frac{\Delta_i}{L}, \quad U_i \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1),$$

and write

$$M = U_{(n)}, \quad c = \frac{L}{v}.$$

Then $T = cM$ and

$$\mu = \mathbb{E}[T] = c \frac{n}{n+1}, \quad \sigma^2 = \text{Var}(T) = c^2 \frac{n}{(n+1)^2(n+2)}. \quad (32)$$

For m independent completion times,

$$\hat{\mu}_m = \frac{1}{m} \sum_{j=1}^m T_j$$

is unbiased, with exact variance and standard error

$$\text{Var}(\hat{\mu}_m) = \frac{n}{m(n+1)^2(n+2)} \left(\frac{L}{v} \right)^2, \quad (33)$$

$$\text{SE}(\hat{\mu}_m) = \frac{L}{v} \sqrt{\frac{n}{m(n+1)^2(n+2)}}. \quad (34)$$

The iid central limit theorem gives the usual asymptotic normalization [6].

$$\frac{\sqrt{m}(\hat{\mu}_m - \mu)}{\sigma} \Rightarrow N(0, 1).$$

Exact antithetic calculation

Reflect a generated vector $U = (U_1, \dots, U_n)$ to

$$U^A = (1 - U_1, \dots, 1 - U_n).$$

The reflected completion statistic is

$$M^A = \max_i (1 - U_i) = 1 - U_{(1)},$$

so the normalized antithetic pair average is

$$A = \frac{M + M^A}{2} = \frac{U_{(n)} + 1 - U_{(1)}}{2}. \quad (35)$$

Both marginals have the same law as M , so cA is unbiased for μ .

Theorem 6.1 (Exact antithetic variance). *For the pair average in equation (35),*

$$\text{Corr}(M, M^A) = -\frac{1}{n}, \quad (36)$$

$$\text{Var}(cA) = \left(\frac{L}{v} \right)^2 \frac{n-1}{2(n+1)^2(n+2)}. \quad (37)$$

Proof. For uniform order statistics [2],

$$\text{Var}(U_{(1)}) = \text{Var}(U_{(n)}) = \frac{n}{(n+1)^2(n+2)},$$

and

$$\text{Cov}(U_{(1)}, U_{(n)}) = \frac{1}{(n+1)^2(n+2)}.$$

Since $M^A = 1 - U_{(1)}$, the covariance of M and M^A is the negative of the latter quantity. Dividing by the common variance gives $-1/n$. Substitution into

$$\text{Var}(A) = \frac{1}{4} [\text{Var}(M) + \text{Var}(M^A) + 2 \text{Cov}(M, M^A)]$$

gives equation (37) after multiplication by $(L/v)^2$. $\square$

If m independent base vectors are used, averaging the m pair estimates gives

$$\text{Var}(\hat{\mu}_m^A) = \frac{1}{m} \left(\frac{L}{v} \right)^2 \frac{n-1}{2(n+1)^2(n+2)}.$$

Against crude Monte Carlo using the same number m of independently generated uniform vectors,

$$\frac{\text{Var}(\hat{\mu}_m^A)}{\text{Var}(\hat{\mu}_m)} = \frac{n-1}{2n}. \quad (38)$$

In this problem, M and M^A are obtained in one scan by tracking the maximum and minimum, so no second random vector is required.

A different accounting convention compares one antithetic pair with two independently generated crude functional values. Against $2m$ independent crude realizations,

$$\frac{\text{Var}(\hat{\mu}_m^A)}{\text{Var}(\hat{\mu}_{2m})} = \frac{n-1}{n}. \quad (39)$$

For $n = 50$, the two ratios are 0.49 and 0.98. These comparisons answer different cost questions: the first fixes random-vector generation, while the second fixes the number of marginal functional values. Neither is a complete wall-clock model, since random-number generation and min/max updates need not have equal cost. The exact formulas make that accounting choice explicit rather than folding it into the variance-reduction claim. Antithetic variates are standard; here their performance is fully identifiable because the completion-time law is available in closed form [4].

A fixed-seed verification script used in the computational audit samples the normalized directed-distance vectors directly and checks equations (32), (36), (38), and (39) numerically.

7 Computational Verification and Companion Visualization

The reduction in equation (1) gives an exact reference computation for deterministic instances: directed exit times require only a scan of the initial positions and directions, followed by sorting if the full chronological sequence is required. The manuscript’s implementation was checked independently against an event-driven collision simulator using exact rational arithmetic. Seeded tests across system sizes through $n = 8$ matched both the full exit-time multiset and the binary-inversion collision count. Exhaustive enumeration of all direction words through $n = 11$ reproduced equations (7), (8), and the q -binomial enumeration in equation (5). Separate numerical checks covered the reflection criterion, the general completion-time law, the terminal Gamma limit, and the antithetic formulas in Section 6.

A companion browser visualization¹ is designed as an expository rendering of the mathematics rather than as data. Randomized ants use the pass-through representation, while a colored benchmark pair displays an explicit collision and reversal. The benchmark includes an inward-facing endpoint trajectory, so the displayed hybrid run realizes the sharp deterministic bound L/v and should not be interpreted as a draw from the pure iid-uniform model. Pixel dimensions and wall-clock animation speed merely rescale the dimensionless quantities x/L and vt/L .

8 Conclusion

The classical relabeling invariant turns the ants-on-a-stick system from a collision-tracking problem into a finite exit problem on directed initial distances. In deterministic form, that reduction yields the exact exit-time multiset, the sharp bound L/v , an optimal $\Theta(n)$ completion-time algorithm, and a comparison-optimal $\Theta(n \log n)$ procedure for the chronological exit sequence. The physical collision

¹Available at <https://arian.ms/work.html#ants-work> under “Ants on a Stick.”

burden remains nontrivial: it is a binary inversion statistic, can be quadratic in n , and under fair directions has an exact q -binomial mixture law with explicit first two moments.

For random initial positions, midpoint reflection symmetry is the exact condition under which every independent joint direction law preserves the iid directed-distance law. Uniform positions then expose the complete exit process through classical order statistics: Beta marginals, Dirichlet spacings, a binomial remaining-particle count with pure-death transitions, and Gamma limits for the final fixed-rank gaps. Outside midpoint symmetry, the general completion-time formula makes the role of the initial direction count explicit.

These closed forms also provide a controlled simulation benchmark. The crude completion-time estimator and its reflected antithetic pair have exact finite-sample variance formulas, making both the benefit and the cost convention transparent. Taken together, these results show how a classical relabeling invariant organizes the deterministic, probabilistic, and computational structure of the finite exit system.

References

- [1] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to Algorithms*. MIT Press, Cambridge, MA, 4 edition, 2022. ISBN 9780262046305.
- [2] Herbert A. David and Haikady N. Nagaraja. *Order Statistics*. Wiley Series in Probability and Statistics. John Wiley & Sons, 3 edition, 2003. ISBN 9780471389262. doi: 10.1002/0471722162.
- [3] Paul Embrechts, Claudia Klüppelberg, and Thomas Mikosch. *Modelling Extremal Events for Insurance and Finance*, volume 33 of *Stochastic Modelling and Applied Probability*. Springer, Berlin, 1997. doi: 10.1007/978-3-642-33483-2.
- [4] Paul Glasserman. *Monte Carlo Methods in Financial Engineering*, volume 53 of *Stochastic Modelling and Applied Probability*. Springer, New York, 2004. ISBN 9780387004518. doi: 10.1007/978-0-387-21617-1.
- [5] Google. Ants on a stick. Google Kick Start 2022, Round C, Google Coding Competitions Archive, 2022. Archived problem statement and analysis available at https://github.com/google/coding-competitions-archive/tree/main/kickstart/2022/round_c/ants_on_a_stick.
- [6] John A. Gubner. *Probability and Random Processes for Electrical and Computer Engineers*. Cambridge University Press, 2006. ISBN 9780521864701. doi: 10.1017/CBO9780511813610.
- [7] Theodore E. Harris. Diffusion with “collisions” between particles. *Journal of Applied Probability*, 2(2):323–338, 1965. doi: 10.2307/3212197.
- [8] Svante Janson. Generalized galois numbers, inversions, lattice paths, ferrers diagrams and limit theorems. *The Electronic Journal of Combinatorics*, 19(3):P34, 2012. doi: 10.37236/2188.
- [9] D. W. Jepsen. Dynamics of a simple many-body system of hard rods. *Journal of Mathematical Physics*, 6(3):405–413, 1965. doi: 10.1063/1.1704288.
- [10] Artem Ryabov. Single-file diffusion in an interval: First passage properties. *The Journal of Chemical Physics*, 138(15):154104, 2013. doi: 10.1063/1.4801326.
- [11] Sanjib Sabhapandit, Ioana Bena, and Satya N. Majumdar. Statistics of the total number of collisions and the ordering time in a freely expanding hard-point gas. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(05):P05012, 2008. doi: 10.1088/1742-5468/2008/05/P05012.
- [12] Francis E. Su et al. Ants on a stick. Math Fun Facts, Harvey Mudd College, n.d. Available at <https://math.hmc.edu/funfacts/ants-on-a-stick/>.
- [13] Peter Winkler. *Mathematical Mind-Benders*. A K Peters, 2007. ISBN 9781568813363.